SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 35%

Duration: 1 Hour

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario Based Assignment

Code of Conduct:

*I(**Sandesh S/192521078**) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Sandesh S

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty

ii. Explain how A* would handle the same situation, including how it balances cost and heuristic

iii. Analyze the impact of heuristic accuracy on both algorithms

iv. Discuss how dynamic changes (blocked paths) affect search performance

v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. **Scenario**: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

i. Explain how Hill Climbing would work in this scenario and identify its limitations

ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation

iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations

iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. **Scenario:** An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

i. Explain why this problem requires an online search agent instead of offline search

ii. Analyze the challenges of partial observability and dynamic environments

iii. Describe how the rover builds and updates its internal model

iv. Compare online search with uninformed and informed search strategies

v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

i. Clearly define variables, domains, and constraints with explanation

ii. Explain how backtracking search works in this scenario

iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency

iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must complete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

i. Explain how the Minimax algorithm models this problem

ii. Analyze the computational challenges of large game trees

iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning

iv. Design an evaluation function and justify the factors considered

v. Recommend a strategy for balancing optimality and real-time performance

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

SCENARIO-BASED ASSIGNMENT

SCENARIO 1: AI-POWERED DRONES FOR EMERGENCY MEDICINE DELIVERY

i. Explain how Greedy Best-First Search behaves in this scenario and discuss its strengths and weaknesses under uncertainty.

Greedy Best-First Search selects the path that appears closest to the destination using a heuristic value.

Strengths:

- Finds promising routes quickly.
- Reduces search time during emergencies.
- Uses straight-line distance to estimate the nearest path.

Weaknesses:

- May select a blocked road.
- Does not consider the complete travel cost.
- May produce unsafe or longer routes when weather changes.

ii. Explain how A* would handle the same situation, including how it balances cost and heuristic.

A* selects the route using both the actual cost and the estimated cost to the destination.

Formula: **f(n) = g(n) + h(n)**

- g(n) represents the actual cost from the starting point.
- h(n) represents the estimated cost to the destination.
- f(n) represents the total estimated cost.

The drone considers distance, weather, blocked paths, and travel time to find a suitable route. It can update its route when new information becomes available.

iii. Analyze the impact of heuristic accuracy on both algorithms.

A good heuristic helps both algorithms find better routes quickly. An accurate distance estimate helps Greedy Best-First Search select promising directions. In A*, an accurate heuristic improves search efficiency and can help find the optimal route when the heuristic does not overestimate the remaining cost. A poor heuristic may cause delays and unnecessary exploration.

iv. Discuss how dynamic changes (blocked paths) affect search performance.

Blocked roads and changing weather can make the original route unsuitable. The drone must update its map and calculate a new route. These changes increase search time and travel cost. Frequent updates help the drone avoid obstacles and deliver medicines safely.

v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety.

A* is more suitable because it considers both actual travel cost and estimated remaining cost. It can find a minimum-cost route when the heuristic and search conditions are appropriate. With real-time updates and safety checks, it can respond to blocked paths and changing weather. For urgent deliveries, the drone should prioritize safety and select a safe route with minimum delay.

SCENARIO 2: AI-BASED TRAFFIC SIGNAL OPTIMIZATION

Formulate this as an optimization problem and explain why traditional search is inefficient.

The objective is to minimize total waiting time, fuel consumption, and traffic congestion by adjusting traffic signal timings.

The system must consider traffic flow, vehicle queues, signal timings, and pedestrian crossing requirements. Traditional search methods may become inefficient because there are many intersections and possible signal combinations. Traffic conditions also change continuously.

i. Explain how Hill Climbing would work in this scenario and identify its limitations.

Hill Climbing starts with an initial set of signal timings and makes small changes to improve traffic flow. It selects a better neighbouring solution and repeats the process until no improvement is found.

Limitations:

- May get stuck in a local maximum or minimum.
- May stop at a plateau.
- May struggle with complex traffic patterns.
- Does not always find the global best solution.

ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation.

- **Local maximum/minimum:** The system finds a solution that is better than nearby solutions but not the best overall.

- **Plateau:** Several signal timing combinations have the same performance, so the system cannot easily decide which direction to move.
- **Ridge:** The best improvement requires changing multiple signal timings together, but individual changes may not improve traffic flow.

These problems can prevent the system from finding the best traffic signal arrangement.

iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations.

Simulated Annealing: It sometimes accepts a worse solution to escape local optima. As the search progresses, it becomes less likely to accept worse solutions.

Genetic Algorithm: It creates a population of possible signal timing solutions. Selection, crossover, and mutation produce new solutions, allowing the system to explore different combinations.

Both methods can explore a larger solution space than basic Hill Climbing.

iv. Recommend the best approach and justify your choice based on scalability and adaptability.

A Genetic Algorithm is a suitable choice because it can optimize many signal timings simultaneously. It explores different combinations and can adapt to changing traffic conditions by generating new solutions. For a large smart city, it can be combined with real-time traffic data and safety constraints to improve overall traffic flow.

SCENARIO 3: AUTONOMOUS ROVER ON MARS

i. Explain why this problem requires an online search agent instead of offline search.

An online search agent makes decisions while exploring the environment. The Mars rover does not have a complete map of the terrain before starting. It must discover obstacles and safe paths using its sensors. Therefore, online search helps it adapt to unknown conditions.

ii. Analyze the challenges of partial observability and dynamic environments.

Partial observability: The rover cannot observe the entire environment at once. Its sensors provide only limited information about nearby areas.

Dynamic environment: New obstacles, changing terrain conditions, or unexpected events may require the rover to change its route.

These challenges make navigation difficult and require continuous observation and decision-making.

iii. Describe how the rover builds and updates its internal model.

The rover collects information using cameras and other sensors. It identifies obstacles, safe areas, and possible routes. It stores this information in an internal map. When new information is received, the rover updates the map and changes its plan if necessary.

iv. Compare online search with uninformed and informed search strategies.

- **Online search:** Explores the environment and makes decisions using newly received information.
- **Uninformed search:** Searches without using additional knowledge about how close a state is to the goal.
- **Informed search:** Uses heuristic information to guide the search toward the goal.

Online search is useful when the environment is unknown. Informed search can improve route selection when useful heuristic information is available.

v. Justify the most suitable approach considering uncertainty, adaptability, and safety.

An online search agent combined with informed search is the most suitable approach. It can use sensor data to explore unknown terrain and heuristic information to choose promising routes. It can update its map when obstacles are discovered. Safety rules must prevent the rover from entering dangerous areas and help it reach its destination reliably.

SCENARIO 4: AI-BASED EXAM TIMETABLE GENERATION

Formulate the problem as a Constraint Satisfaction Problem (CSP).

A Constraint Satisfaction Problem consists of variables, domains, and constraints. The university must assign exams to suitable time slots and halls without violating scheduling rules.

i. Clearly define variables, domains, and constraints with explanation.

Variables:

- Each exam to be scheduled.
- Time slot assigned to each exam.
- Exam hall assigned to each exam.

Domains:

- Available examination time slots.
- Available examination halls.

- Available faculty members.

Constraints:

- A student must not have two exams at the same time.
- The number of students must not exceed hall capacity.
- Faculty members must be available.
- Certain subjects must be scheduled before others.
- Special accommodations must be provided where required.

These constraints help produce a valid timetable.

ii. Explain how backtracking search works in this scenario.

Backtracking assigns a time slot and hall to an exam one at a time. After each assignment, it checks whether all constraints are satisfied. If a conflict occurs, it removes the assignment and tries another option. The process continues until a valid timetable is found.

iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency.

Forward checking removes invalid choices from the domains of unassigned exams after each assignment. For example, if a student has an exam at 10 AM, other exams taken by that student cannot be scheduled at the same time.

This reduces unnecessary searching and helps detect conflicts early.

iv. Analyze real-world challenges and justify the best strategy for scalability.

Real-world challenges include a large number of students, limited halls, faculty availability, special accommodations, and changing schedules.

A combination of backtracking, forward checking, and suitable variable-ordering techniques is effective. Choosing the most constrained exam first can reduce the search space. For large universities, optimization techniques can also be used to improve scalability.

SCENARIO 5: STRATEGIC GAME AI FOR REAL-TIME DECISION-MAKING

i. Explain how the Minimax algorithm models this problem.

Minimax is used in two-player games where one player tries to maximize the score and the opponent tries to minimize it.

- The AI represents possible moves using a game tree.
- MAX selects the move with the highest value.

- MIN selects the move with the lowest value.
- The algorithm evaluates possible future game states.
- It chooses the move that gives the best outcome against the opponent.

ii. Analyze the computational challenges of large game trees.

Large game trees contain many possible moves and future states. The number of states increases rapidly as the search depth grows. Evaluating every possible move takes considerable time and memory. This makes it difficult for the AI to make decisions within real-time limits.

iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning.

Alpha-Beta pruning improves Minimax by removing branches that cannot affect the final decision.

- **Alpha:** The best value MAX can guarantee so far.
- **Beta:** The best value MIN can guarantee so far.
- When alpha becomes greater than or equal to beta, the remaining branches can be skipped.
- This reduces the number of states that need to be evaluated.
- It produces the same optimal decision as Minimax when the same search depth and evaluation rules are used.

Therefore, Alpha-Beta pruning saves computation time and helps the AI make faster decisions.

iv. Design an evaluation function and justify the factors considered.

An evaluation function estimates how good a game state is when the AI cannot search all the way to the end of the game.

Example:

Evaluation Score = (10 × AI Score) − (10 × Opponent Score) + Position Bonus

The factors include:

- AI score and opponent score.
- Strength of the current position.
- Available moves and future opportunities.
- Immediate threats and possible winning moves.

A good evaluation function helps the AI choose promising moves when the search is limited.

v. Recommend a strategy for balancing optimality and real-time performance.

The best strategy is to combine Minimax with Alpha-Beta pruning and a depth limit. Move ordering can help the algorithm examine promising moves first. An evaluation function estimates the value of states at the depth limit. Iterative deepening can increase the search depth gradually while respecting the time limit. This approach provides strong decisions while maintaining real-time performance.

CONCLUSION

Artificial Intelligence search and optimization algorithms solve different real-world problems. Greedy Best-First Search and A* help with route planning, while Hill Climbing, Simulated Annealing, and Genetic Algorithms support optimization. Online search helps autonomous agents explore unknown environments. Constraint Satisfaction Problems support timetable generation, and Minimax with Alpha-Beta pruning enables efficient game decisions. The best algorithm depends on the problem, available information, time limits, and safety requirements.